P2 Chapter 10: Numerical Methods

Locating Roots

Why do we need numerical methods?

Finding the root of a function $f(x)$ is to:
solve the equation $f(x) = 0$
(i.e. the inputs such that the output of the function is 0)

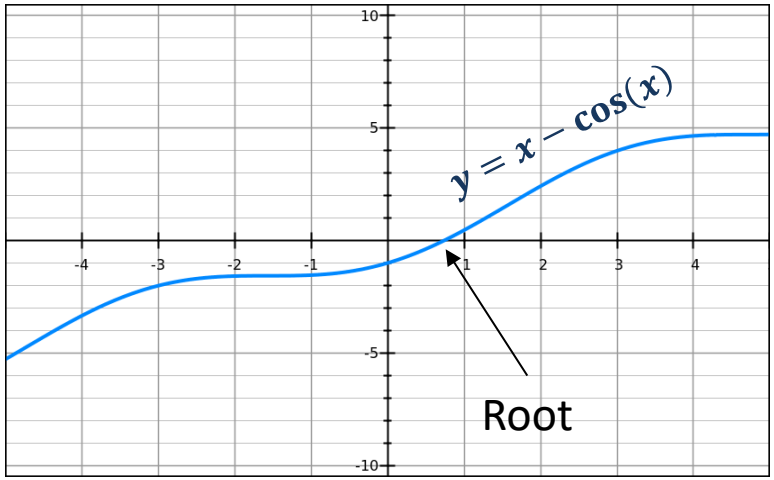
Terminology Pedantry: We often say "*Find the roots of* $x^2 + x = 0$", i.e. an <u>**equation**</u> in the form $f(x) = 0$, which really means "*Find the roots of the function* $f(x)$ where $f(x) = x^2 + x$". But we <u>wouldn't</u> say "Find the roots of $x^2 - x = 6$", but "Find the **solutions** of…". So to find roots is to find the x such that the '<u>**output**</u>', <u>**or one side of the equation**</u>, <u>**is 0**</u>.

However, for some functions, the 'exact' root is either complicated and difficult to calculate:

$$\boldsymbol{x^3 + 2x^2 - 3x + 4 = 0} \quad \Rightarrow \quad x = \frac{1}{3}\left(-2 - \frac{13}{\sqrt[3]{89 - 6\sqrt{159}}} - \sqrt[3]{89 - 6\sqrt{159}}\right)$$

or there's no 'algebraic' expression at all! (involving roots, logs, sin, cos, etc.)

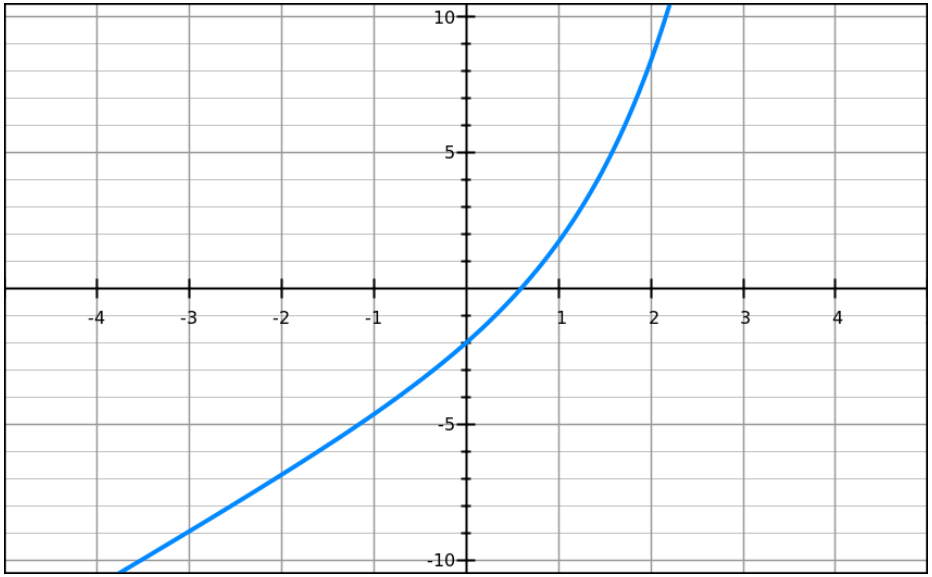
$$\boldsymbol{x - \cos(x) = 0} \quad \Rightarrow \quad \textbf{Exact solution not expressible ☹}$$



But there are a variety of 'numerical methods' which get progressively better solutions to an equation in the form $f(x) = 0$.
You have already seen 'iteration' at GCSE as one such method.

Proving a solution lies in a range

Show that $f(x) = e^x + 2x - 3$ has a root between $x = 0.5$ and $x = 0.6$



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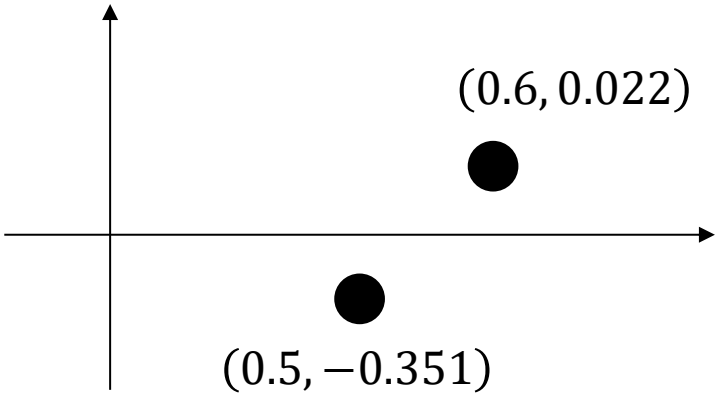

If the y value goes from negative to positive or vice versa, then clearly the y values must pass 0 somewhere in between.

Exam Tip: The mark scheme lookings for:

1. Finding the function output for the two values.
2. Referring to a 'change in sign'.

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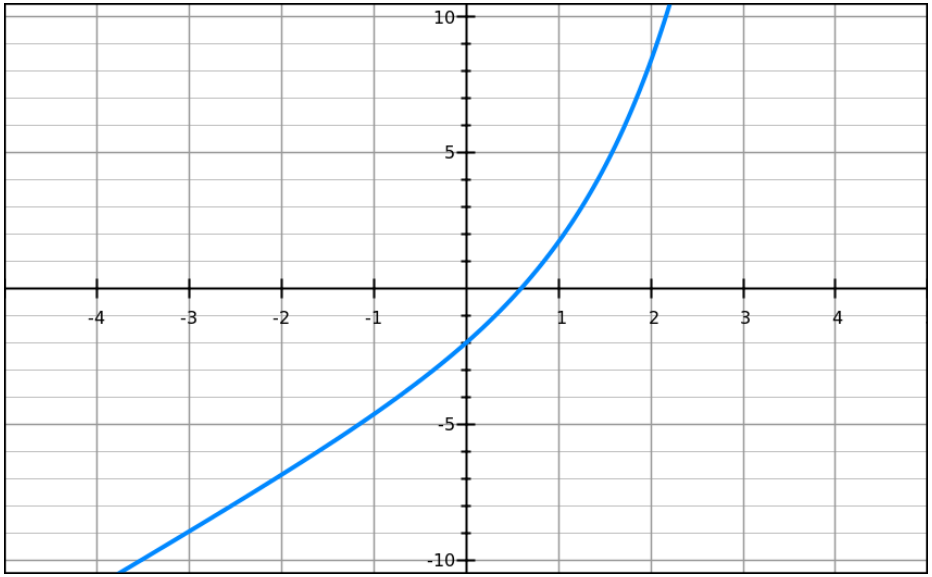
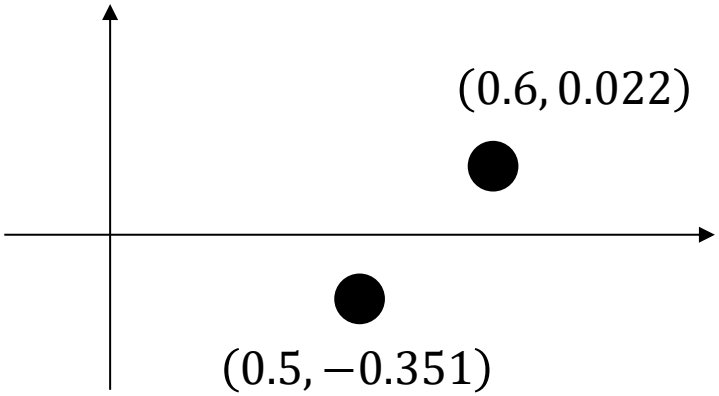


$$f(0.5) = -0.351$$
$$f(0.6) = 0.022 \ldots$$

There is a **<u>change in sign</u>**, and $\boldsymbol{f(x)}$ **is continuous**, so root must lie between 0.5 and 0.6



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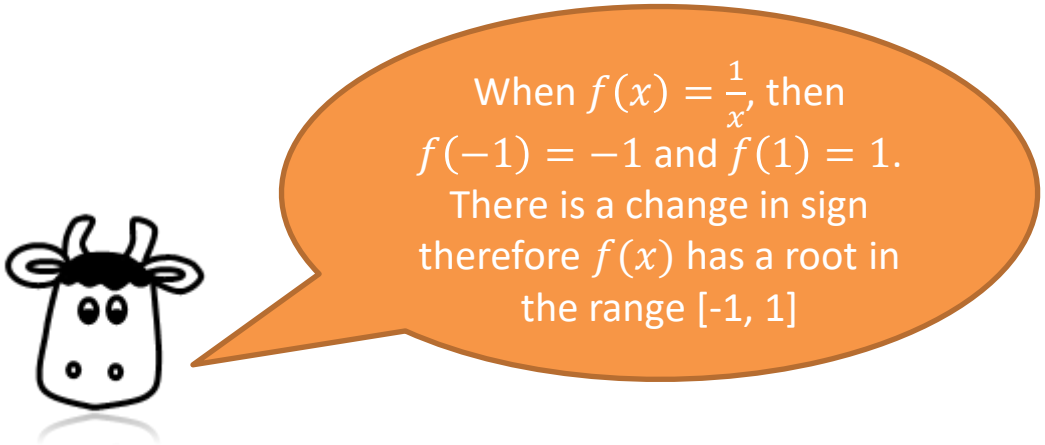
Exam Tip: The mark scheme lookings for:

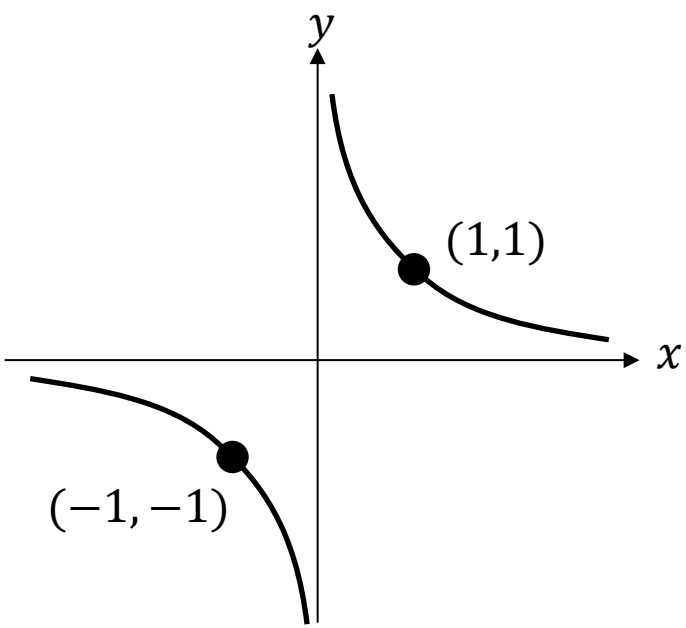
1. Finding the function output for the two values.
2. Referring to a 'change in sign'.

...but only if the function is continuous

Stupid Steve says:

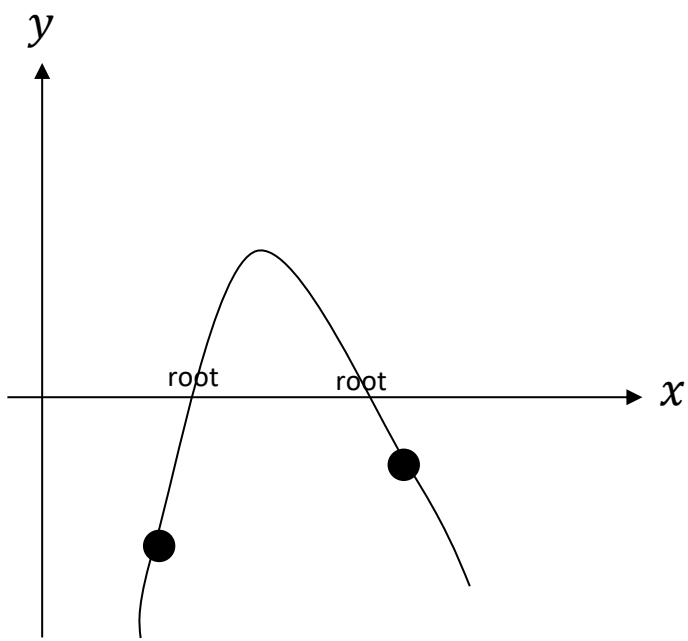


Why is Steve wrong?



A function is **continuous** if the line **does not 'jump'**. A root is only guaranteed with a sign change if the function is continuous, as otherwise the line can skip past 0 (in this case due to a vertical asymptote.

…and no sign change doesn't mean there isn't a root



Beware! Just because there isn't a sign change, doesn't mean there's no root in that interval.

The sign change method fails to detect a root if there were an **even number of roots** in that interval.

Proving a solution to a given accuracy

Edexcel C3 Jan 2013

$$g(x) = e^{x-1} + x - 6$$

The root of $g(x) = 0$ is α.

(c) By choosing a suitable interval, show that $\alpha = 2.307$ correct to 3 decimal places.

(3)

Thinking back to lower school, if the root 2.307 is correct to 3dp, what's the smallest and greatest value it could be?

Smallest: ?

Greatest: ?

If there was a sign change between $x = 2.3065$ and $x = 2.3075$, then we know $2.3065 < \alpha < 2.3075$. But if the value is in this range, then we know it is 2.307 to 3 decimal places!

? What do we write in exam…

Proving a solution to a given accuracy

Edexcel C3 Jan 2013

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Greatest: 2.3075

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$$\boldsymbol{f(2.3065) = -0.000275}$$
$$\boldsymbol{f(2.3075) = 0.00442}$$

There is a change in sign in the interval $\boldsymbol{(2.3065, 2.3075)}$

$$\boldsymbol{\therefore 2.3065 < \alpha < 2.3075}$$
$$\boldsymbol{\therefore \alpha = 2.307 \ \ to\ 3dp}$$

Example

[Textbook] (a) Using the same axes, sketch the graphs of $y = \ln x$ and $y = \frac{1}{x}$. Explain how your diagrams shows that the function $y = \ln(x) - \frac{1}{x}$ has only one root.
(b) Show that this root lies in the interval $1.7 < x < 1.8$
(c) Given that the root of $f(x)$ is α, show that $\alpha = 1.763$ correct to 3 decimal places.

a

?

b

?

c

?

Example

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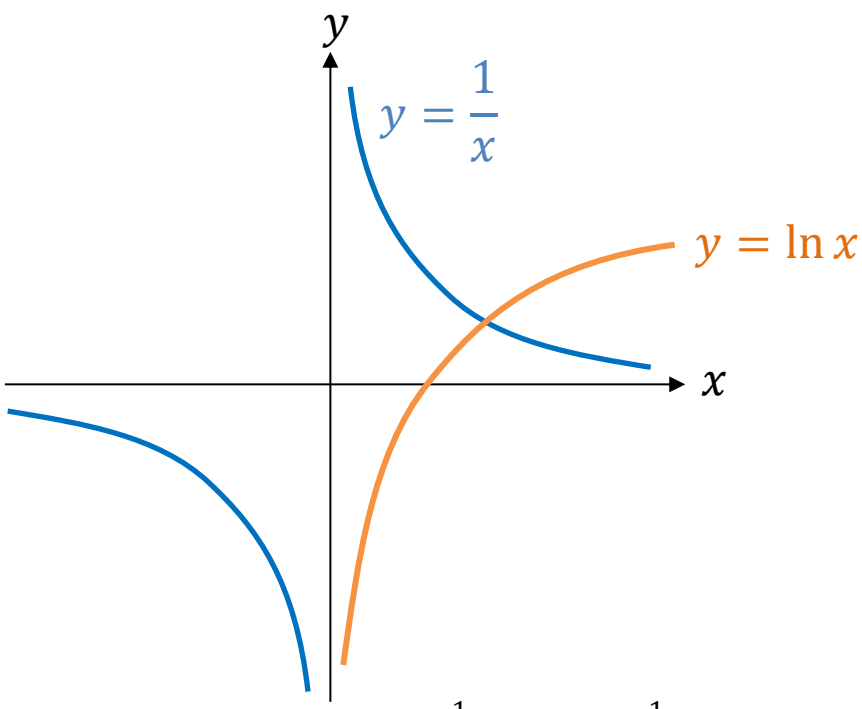

The lines intersect where $\ln x = \frac{1}{x} \rightarrow \ln x - \frac{1}{x} = 0$

Thus the roots of $\ln x - \frac{1}{x} = 0$ correspond to the points of intersection, and there is only one point of intersection on the graph.

b

$$f(x) = \ln x - \frac{1}{x}$$
$$f(1.7) = -0.0576 \ldots$$
$$f(1.8) = 0.0322 \ldots$$

There is a change of sign between 1.7 and 1.8, so there is a root in the interval $1.7 < x < 1.8$

c

$$f(1.17625) = -0.00064 < 0$$
$$f(1.7635) = 0.00024 > 0$$

There is a change of sign in the interval $(1.7625, 1.7635)$ so $1.7625 < \alpha < 1.7635$, so $\alpha = 1.763$ correct to 3dp.

Exercise 10.1

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Homework Exercise

1 Show that each of these functions has at least one root in the given interval.

a $f(x) = x^3 - x + 5, -2 < x < -1$

b $f(x) = x^2 - \sqrt{x} - 10, 3 < x < 4$

c $f(x) = x^3 - \frac{1}{x} - 2, -0.5 < x < -0.2$

d $f(x) = e^x - \ln x - 5, 1.65 < x < 1.75$

2 $f(x) = 3 + x^2 - x^3$

a Show that the equation $f(x) = 0$ has a root, α, in the interval [1.8, 1.9]. **(2 marks)**

b By considering a change of sign of $f(x)$ in a suitable interval, verify that $\alpha = 1.864$ correct to 3 decimal places. **(3 marks)**

3 $h(x) = \sqrt[3]{x} - \cos x - 1$, where x is in radians.

a Show that the equation $h(x) = 0$ has a root, α, between $x = 1.4$ and $x = 1.5$. **(2 marks)**

b By choosing a suitable interval, show that $\alpha = 1.441$ is correct to 3 decimal places. **(3 marks)**

4 $f(x) = \sin x - \ln x, x > 0$, where x is in radians.

a Show that $f(x) = 0$ has a root, α, in the interval [2.2, 2.3]. **(2 marks)**

b By considering a change of sign of $f(x)$ in a suitable interval, verify that $\alpha = 2.219$ correct to 3 decimal places. **(3 marks)**

5 $f(x) = 2 + \tan x, 0 < x < \pi$, where x is in radians.

a Show that $f(x)$ changes sign in the interval [1.5, 1.6].

b State with a reason whether $f(x)$ has a root in the interval [1.5, 1.6].

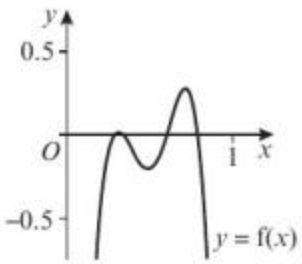
Homework Exercise

6 A student observes that the function $f(x) = \frac{1}{x} + 2$, $x \neq 0$, has a change of sign on the interval $[-1, 1]$. The student writes:

> $y = f(x)$ has a vertical asymptote within this interval so even though there is a change of sign, $f(x)$ has no roots in this interval.

By means of a sketch, or otherwise, explain why the student is incorrect.

7 $f(x) = (105x^3 - 128x^2 + 49x - 6)\cos 2x$, where x is in radians.
The diagram shows a sketch of $y = f(x)$.



a Calculate $f(0.2)$ and $f(0.8)$.

b Use your answer to part **a** to make a conclusion about the number of roots of $f(x)$ in the interval $0.2 < x < 0.8$.

c Further calculate $f(0.3)$, $f(0.4)$, $f(0.5)$, $f(0.6)$ and $f(0.7)$.

d Use your answers to parts **a** and **c** to make an improved conclusion about the number of roots of $f(x)$ in the interval $0.2 < x < 0.8$.

8 **a** Using the same axes, sketch the graphs of $y = e^{-x}$ and $y = x^2$.

b Explain why the function $f(x) = e^{-x} - x^2$ has only one root.

c Show that the function $f(x) = e^{-x} - x^2$ has a root between $x = 0.70$ and $x = 0.71$.

9 **a** On the same axes, sketch the graphs of $y = \ln x$ and $y = e^x - 4$.

b Write down the number of roots of the equation $\ln x = e^x - 4$.

c Show that the equation $\ln x = e^x - 4$ has a root in the interval $(1.4, 1.5)$.

Homework Exercise

10 $h(x) = \sin 2x + e^{4x}$

a Show that there is a stationary point, α, of $y = h(x)$ in the interval $-0.9 < x < -0.8$. **(4 marks)**

b By considering the change of sign of $h'(x)$ in a suitable interval, verify that $\alpha = -0.823$ correct to 3 decimal places. **(2 marks)**

11 **a** On the same axes, sketch the graphs of $y = \sqrt{x}$ and $y = \frac{2}{x}$ **(2 marks)**

b With reference to your sketch, explain why the equation $\sqrt{x} = \frac{2}{x}$ has exactly one real root. **(1 mark)**

c Given that $f(x) = \sqrt{x} - \frac{2}{x}$, show that the equation $f(x) = 0$ has a root r, where $1 < r < 2$. **(2 marks)**

d Show that the equation $\sqrt{x} = \frac{2}{x}$ may be written in the form $x^p = q$, where p and q are integers to be found. **(2 marks)**

e Hence write down the exact value of the root of the equation $\sqrt{x} - \frac{2}{x} = 0$. **(1 mark)**

12 $f(x) = x^4 - 21x - 18$

a Show that there is a root of the equation $f(x) = 0$ in the interval $[-0.9, -0.8]$. **(3 marks)**

b Find the coordinates of any stationary points on the graph $y = f(x)$. **(3 marks)**

c Given that $f(x) = (x - 3)(x^3 + ax^2 + bx + c)$, find the values of the constants a, b and c. **(3 marks)**

d Sketch the graph of $y = f(x)$. **(3 marks)**

Homework Answers

1 **a** $f(-2) = -1 < 0$, $f(-1) = 5 > 0$. Sign change implies root.

b $f(3) = -2.732 < 0$, $f(4) = 4 > 0$. Sign change implies root.

c $f(-0.5) = -0.125 < 0$, $f(-0.2) = 2.992 > 0$. Sign change implies root.

d $f(1.65) = -0.294 < 0$, $f(1.75) = 0.195 > 0$. Sign change implies root.

2 **a** $f(1.8) = 0.408 > 0$, $f(1.9) = -0.249$. Sign change implies root.

b $f(1.8635) = 0.00138\ldots > 0$, $f(1.8645) = -0.00531\ldots < 0$. Sign change implies root.

3 **a** $h(1.4) = -0.0512\ldots < 0$, $h(1.5) = 0.0739\ldots. > 0$. Sign change implies root.

b $h(1.4405) = -0.00055\ldots < 0$, $h(1.4415) = 0.00069\ldots > 0$. Sign change implies root.

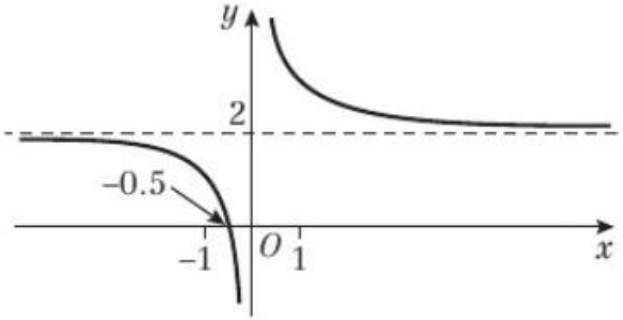
4 **a** $f(2.2) = 0.020 > 0$, $f(2.3) = -0.087$. Sign change implies root.

b $f(2.2185) = 0.00064\ldots > 0$, $f(2.2195) = -0.00041\ldots < 0$. There is a sign change in the interval $2.2185 < x < 2.2195$, so $\alpha = 2.219$ correct to 3 decimal places.

5 **a** $f(1.5) = 16.10\ldots > 0$, $f(1.6) = -32.2\ldots < 0$. Sign change implies root.

b There is an asymptote in the graph of $y = f(x)$ at $x = \frac{\pi}{2} \approx 1.57$. So there is not a root in this interval.

6



Alternatively: $\frac{1}{x} + 2 = 0 \Rightarrow \frac{1}{x} = -2 \Rightarrow x = -\frac{1}{2}$

7 **a** $f(0.2) = -0.4421\ldots$, $f(0.8) = -0.1471\ldots$

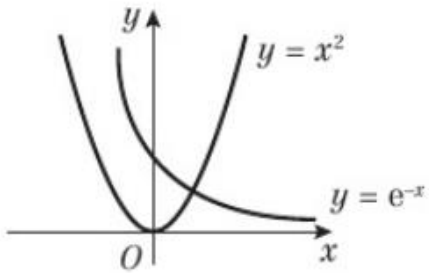
b There are either no roots or an even numbers of roots in the interval $0.2 < x < 0.8$.

c $f(0.3) = 0.01238\ldots > 0$, $f(0.4) = -0.1114\ldots < 0$, $f(0.5) = -0.2026\ldots < 0$, $f(0.6) = 0$, $f(0.7) = -0.2710\ldots > 0$

d There exists at least one root in the interval $0.2 < x < 0.3$, $0.3 < x < 0.4$ and $0.7 < x < 0.8$. Additionally $x = 0.6$ is a root. Therefore there are at least four roots in the interval $0.2 < x < 0.8$.

Homework Answers

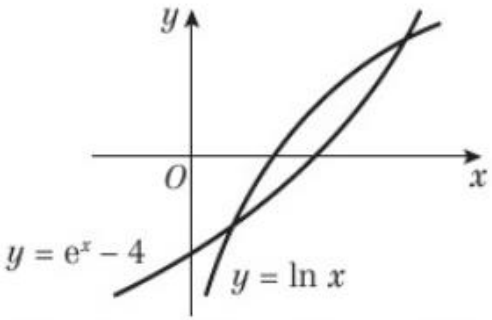
8 **a**



b One point of intersection, so one root.

c $f(0.7) = 0.0065\ldots > 0$, $f(0.71) = -0.0124\ldots < 0$.
Sign change implies root.

9 **a**



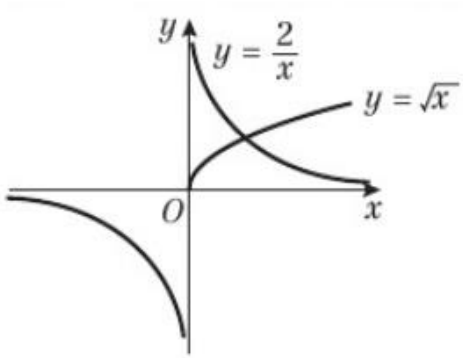
b 2

c $f(x) = \ln x - e^x + 4$. $f(1.4) = 0.2812\ldots < 0$,
$f(1.5) = -0.0762\ldots < 0$. Sign change implies root.

10 **a** $h'(x) = 2\cos 2x + 4e^{4x}$. $h'(-0.9) = -0.3451\ldots < 0$.
$h'(-0.8) = 0.1046\ldots > 0$. Sign change implies slope changes from decreasing to increasing over interval, which implies turning point.

b $h'(-0.8235) = -0.003839\ldots. < 0$,
$h'(-0.8225) = 0.00074\ldots > 0$. Sign change implies α lies in the range $-0.8235 < \alpha < -0.8225$,
so $\alpha = -0.823$ correct to 3 decimal places.

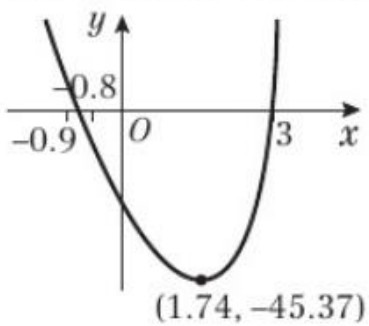
11 **a**



b 1 point of intersection ⇒ 1 root

c $f(1) = -1$, $f(2) = 0.414\ldots$ **d** $p = 3$, $q = 4$ **e** $4^{\frac{1}{3}}$

12 **a** $f(-0.9) = 1.5561 > 0$, $f(-0.8) = -0.7904 < 0$.
There is a change of sign in the interval $[-0.9, -0.8]$, so there is at least one root in this interval.

b (1.74, −45.37) to 2 d.p. **c** $a = 3$, $b = 9$ and $c = 6$

d